

# The question

Every contact number MarFold publishes comes from a 554-protein eval set whose FoldBench half is the **first 100 rows** of `monomer_protein.csv` — chosen before we understood how much of the eval set our training corpora contained (#213, #225). #232 then trained the #199 recipe from scratch on corpora decontaminated against **all** of FoldBench, which makes the other 234 monomers a real held-out test set for those checkpoints for the first time.

So: cut FoldBench's 334 monomers into three sets, score them, and ask whether the set we have been reporting on was telling us the truth.

- **eval-val (97)** — the natural monomers inside the historical FoldBench-100. - **eval-test (217)** — every other natural FoldBench monomer, never scored here. - **eval-denovo (19)** — every de novo designed FoldBench monomer.

Every protein carries a viral flag, because #241 showed the two strata rank models differently.

# What was checked before anything was scored

**\*\*Decontamination, five links, all verified.\*\*** All 334 monomer sequences are in #225's decontamination reference byte-for-byte; 131,180 training rows match one of them at the applied rule ( $\geq 30$  % identity over  $\geq 50$  % of the shorter sequence) and **\*\*all 131,180 are in the drop list — zero survive\*\***; the published corpora match #225's counts; #232's tokenizer pins those exact counts and those exact bucket prefixes; and both W&B runs read only those caches.

**\*\*What the rule does not cover, priced rather than described.\*\*** At the applied gate the highest surviving identity to any of the 334 is 0.299. Relax the coverage requirement to 40 % and essentially every protein has a surviving training relative at  $\geq 30$  % identity; with no coverage requirement, 65 of them have one at  $\geq 90$  % over a fragment. "Decontaminated at 30 %" means that rule, not "no shared subsequence".

**\*\*Ground truth rebuilt through one path\*\***, with the 126 overlapping frozen units reproduced byte-identically. **\*\*One protein excluded\*\***: `8uxt\_A` (1,596 residues) has no representable contacts-v1 document at an 8,192-token context.

# The evaluation reproduces PR #244

Two of the three checkpoints are the ones #244 scored, and all 97 eval-val proteins are inside #244's universe, so the path is validated against a published reference protein by protein rather than against a tolerance on an aggregate: mean R-precision differs by **\*\*−0.0031\*\*** (m2-p06) and **\*\*−0.0023\*\*** (m1-p02) with per-protein  $r = 0.996$  — inside the 0.0023 spread #204 measured for one unchanged checkpoint. 333 units × 3 checkpoints, **\*\*0 unfinished rollouts\*\***.

The baseline path has its own exact control: re-scoring 12 published proteins through this experiment's rebuilt ground truth reproduces #213's ESMFold and ESMFold2 numbers with **\*\*max absolute difference 0.0\*\***.

# The result: eval-val was not flattering us

All-range R-precision, quoted as **\*\*eval-val (97) / eval-test (217) / eval-denovo (19)\*\***.

The two decontaminated #232 checkpoints: **\*\*m2-p06 0.520 / 0.538 / 0.591\*\*** and **\*\*m1-p02 0.473 / 0.493 / 0.588\*\***. The contaminated reference, #199's cooldown: **\*\*0.589 / 0.613 / 0.619\*\***.

Baselines on the same proteins: Protenix-v2 single-seq 0.263 / 0.265 / 0.835; ESMFold 0.750 / 0.753 / 0.795; ESMFold2 0.802 / 0.792 / 0.864; Protenix-v2 + MSA 0.846 / 0.845 / 0.844. The sequence-KNN null scores 0.584 / 0.582 / 0.066 over the unfiltered corpus and 0.407 / 0.426 / 0.050 over the decontaminated one.

**\*\*Every predictor scores the same or slightly better on the 217 monomers we had never touched\*\***, and the contaminated reference moves the same way and by the same amount as the decontaminated checkpoints: the difference-in-differences is **\*\*−0.006\*\*** (m2-p06) and **\*\*−0.004\*\*** (m1-p02), an order of magnitude inside the noise. H2 is not supported — the historical FoldBench-100 was not inflating #199's score through memorised homologs. H1 and H3 hold: every predictor moves by under 0.03 between the two sets.

# Two things this changes

**\*\*The KNN null is the yardstick, and it is corpus-specific.\*\*** Copying the contacts of the ten nearest training sequences scores 0.582 on eval-test out of the unfiltered corpus and 0.426 out of the decontaminated one — 0.156 of memorisable contact map per protein, over 0.2 for 99 of the 314 natural monomers. Each model clears the null over the corpus it trained on (#232 m2-p06 **\*\*+0.112\*\***, #199 cooldown **\*\*+0.031\*\***) and falls below the null over the richer one. #199's 0.075 lead is therefore not a measurement of what decontamination costs — the runs are not budget-matched, 290,400 vs 145,199 steps — but it bounds how much of any contacts-v1 score is reachable by memorisation.

**\*\*Protenix-v2 single-sequence is not a comparator on natural proteins.\*\*** 0.835 on the 19 designs, 0.265 on the 314 natural monomers. The "parity with Protenix-SS" framing came from an eval set that is three-quarters designed protein; on natural FoldBench monomers both #232 checkpoints beat it by more than 0.27. The real gap is to ESMFold2 (−0.255) and Protenix + MSA (−0.307).

**\*\*Viral proteins are harder for everything except MSA\*\*** — eval-test viral vs non-viral: m2-p06 0.465/0.542, #199 cooldown 0.497/0.621, ESMFold2 0.608/0.804, seq-KNN 0.262/0.602, Protenix + MSA 0.812/0.847. #241's finding survives on new proteins, and the gap tracks reachable homology.

# Use from here

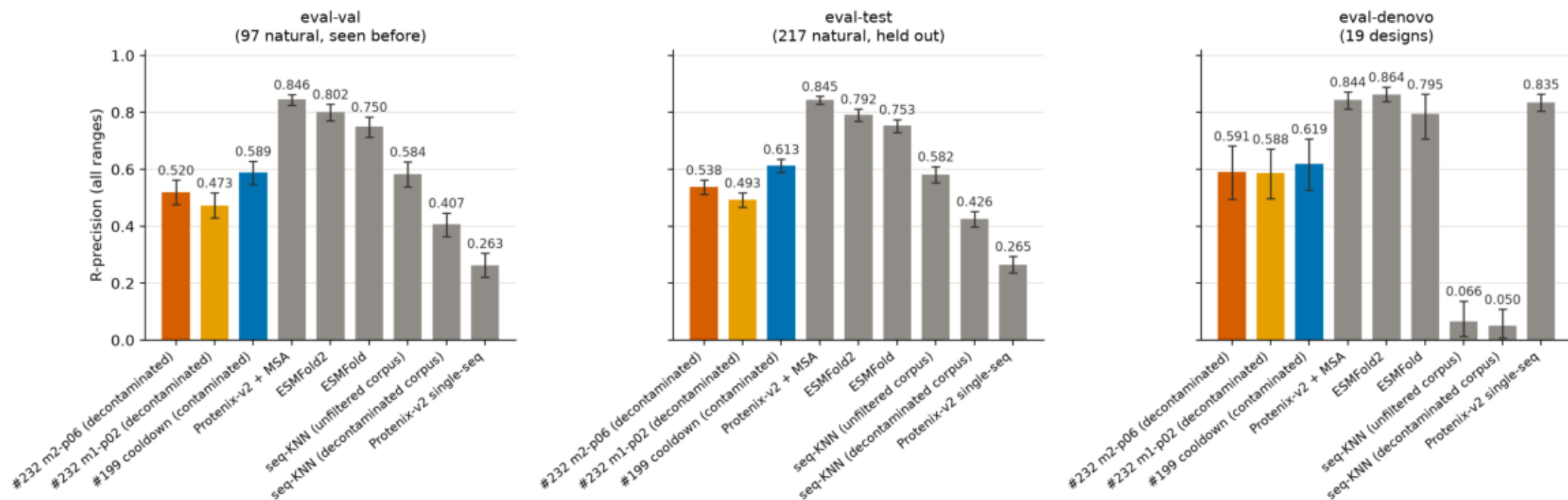
`eval-test` is the default set for a decontaminated-accuracy claim: 217 natural proteins, four times eval2-natural's audited 63, not three-quarters designed, ground truth and all five baselines published. `eval-val` keeps continuity with every published figure. `eval-denovo` keeps designs out of natural-protein means.

Everything is on the bucket under `data/contacts-v1-foldbench-monomers-exp245/`.

# eval\_sets\_scoreboard.png

All-range R-precision per predictor on each of the three FoldBench monomer eval sets, with 95 % bootstrap intervals over proteins. MarFold checkpoints coloured (orange/amber = the decontaminated #232 pair, blue = the contaminated #199 cooldown reference), baselines grey.

Contact R-precision on FoldBench's monomers, split into what we had already scored and what we had not



## val\_vs\_test.png

Each predictor's mean R-precision on eval-val (the 97 natural monomers we have always scored) joined to eval-test (the 217 we never had). Every slope is under 0.03 and the contaminated model's is not steeper than the decontaminated ones', which is the experiment's main result.

